Airline Passenger Satisfaction Analysis

Business Requirement

Business Scenario

An international airline wants to improve its customer experience by understanding the factors that influence passenger satisfaction. Every day, thousands of passengers travel using different classes, travel for various purposes, and experience different levels of service. The airline management wants to identify which services contribute most to customer satisfaction and which areas require improvement.

As a **Junior Data Analyst**, your responsibility is to analyze the airline passenger satisfaction dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify customer satisfaction trends, evaluate service quality, analyze travel patterns, and provide actionable business recommendations.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide meaningful recommendations.

Project Objective

Analyze the airline passenger satisfaction dataset to:

- Understand customer satisfaction levels.
- Analyze passenger demographics and travel behavior.
- Evaluate airline service quality.
- Study the impact of flight delays on customer satisfaction.
- Generate business insights and recommendations to improve passenger experience.

Dataset Description

The dataset contains information about **airline passengers**, including customer demographics, travel details, service ratings, flight delays, and overall satisfaction.

Dataset Features

Column	Description
ID	Unique Passenger Identifier
Gender	Passenger Gender
Customer Type	Loyal or Disloyal Customer
Age	Passenger Age
Type of Travel	Business or Personal Travel
Class	Travel Class (Business, Eco, Eco Plus)
Flight Distance	Distance Travelled
Inflight WiFi Service	Rating of WiFi Service
Departure/Arrival Time Convenient	Rating
Ease of Online Booking	Rating
Gate Location	Rating
Food and Drink	Rating
Online Boarding	Rating
Seat Comfort	Rating
Inflight Entertainment	Rating
On-board Service	Rating
Leg Room Service	Rating
Baggage Handling	Rating
Check-in Service	Rating
Inflight Service	Rating
Cleanliness	Rating
Departure Delay in Minutes	Departure Delay
Arrival Delay in Minutes	Arrival Delay
Satisfaction	Passenger Satisfaction

Note: Ignore the columns **Unnamed: 0** and **id** during analysis, as they are identifier columns and do not contribute to the business analysis.

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
- Load the dataset using Pandas.
- Display the first five records.
- Display the last five records.
- Find the number of rows and columns.
- Display all column names.
- Check the data types of each column.

Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
- Check duplicate records.
- Display summary statistics.
- Find the number of unique customer types.
- Find the number of travel classes.
- Find the number of travel types.
- Find the number of satisfaction categories.

Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
- Handle missing values.
- Remove unnecessary columns (**Unnamed: 0** and **id**).
- Ensure **Age**, **Flight Distance**, **Departure Delay in Minutes**, and **Arrival Delay in Minutes** have the correct data types.
- Remove unnecessary spaces from text columns (if required).

Business Questions

Answer the following questions using Python and Pandas.

Customer Analysis

- What percentage of passengers are satisfied and dissatisfied?
- Which customer type (Loyal or Disloyal) is more satisfied?
- Which gender has higher customer satisfaction?
- Which age group travels the most?

Travel Analysis

- Which travel class has the highest number of passengers?
- Which travel class has the highest customer satisfaction?
- Which type of travel (Business or Personal) has higher satisfaction?
- What is the average flight distance travelled by passengers?

Service Quality Analysis

- Which service receives the highest average rating?
- Which service receives the lowest average rating?
- Compare average ratings for Seat Comfort, Food and Drink, WiFi, Entertainment, Cleanliness, and Online Boarding.
- Which service should the airline improve first?

Flight Delay Analysis

- What is the average departure delay?
- What is the average arrival delay?
- Which passenger group experiences the highest delays?
- Does customer satisfaction decrease with higher flight delays?

Satisfaction Analysis

- Which travel class has the highest number of satisfied passengers?
- Which customer type reports the highest satisfaction?

- Which services receive higher ratings from satisfied passengers?
- Which services receive lower ratings from dissatisfied passengers?

Passenger Preference Analysis

- Which travel purpose is more common among passengers?
- Which travel class is preferred by business travellers?
- Which travel class is preferred by personal travellers?

Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Customer Satisfaction Distribution (Pie Chart)
- Passenger Distribution by Travel Class (Bar Chart)
- Average Service Ratings (Bar Chart)
- Passenger Age Distribution (Histogram)
- Average Flight Delay by Satisfaction Category (Bar Chart)

Optional Chart

- Customer Type Distribution (Pie Chart)

Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which customer segment is the most satisfied?
- Which airline service receives the highest customer ratings?
- Which service requires immediate improvement?
- How do flight delays affect customer satisfaction?
- Which travel class provides the best customer experience?

Business Recommendations

Provide **at least 3 recommendations** for the airline company.

Example recommendations:

- Improve services that receive consistently low customer ratings, such as WiFi or online booking.
- Reduce flight delays to improve passenger satisfaction.
- Enhance customer experience for Economy Class passengers through better onboard services.
- Develop loyalty programs to retain satisfied customers and increase repeat travel.

Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
- Cleaned Dataset (if modified)
- Report
- Business Recommendations (if any)
- 5-minute PowerPoint Presentation

Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	**100**